PANTHEON: Emergent Specialisation and Economic Coordination in a Multi-Agent Cognitive Society

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ABSTRACT

We present PANTHEON, a self-organizing multi-agent cognitive society in which 201 tasks were completed across 200 ticks by a population of identical transformer agents operating under economic pressure. Agents began with identical skill fingerprints and no assigned roles. Through repeated task exposure, reputation feedback, and evolutionary selection, three distinct specialist clusters emerged: code, research, and visual. Coalition formation — driven by a compute-token auction mechanism — achieved a 100% win rate over solo baselines on hard tasks. Mean solution quality stabilised at 0.795 across all task types. The society underwent 2 evolutionary replacement events, with weak agents replaced by offspring of high-performing specialists. All results were produced on a single M4 MacBook without GPU acceleration. This paper was partially generated by the society itself.

INTRODUCTION

Modern multi-agent systems assign roles statically: a planner delegates to a coder who passes output to a critic. This pipeline architecture is brittle, does not adapt, and conflates coordination with intelligence. PANTHEON takes a different approach. Agents begin identical — same architecture, same weights, same token balance. Roles emerge through three forces: task exposure shapes skill fingerprints, reputation routing directs future tasks, and economic pressure eliminates agents that fail to earn compute tokens. The result is a computational society whose collective intelligence exceeds the sum of its parts. Inspired by Generative Agents (Park et al., 2023), CAMEL (Li et al., 2023), nanoGPT (Karpathy, 2022), and EvoPrompting (Chen et al., 2023), PANTHEON contributes: (1) emergent role specialisation without label supervision, (2) a compute-token economy as a coordination mechanism, (3) evolutionary pressure applied to agent cognitive fingerprints, and (4) a shared episodic and semantic memory system enabling retrieval-augmented generation across the society.

METHODS

The substrate layer initialises 10 agents with 100 compute tokens each. Each agent maintains a 128-dimensional skill fingerprint, updated after every task via a weighted pull toward the task-type attractor in fingerprint space. Tasks are drawn from a bank of 20 problems spanning three types (code, research, visual) across three difficulty tiers (easy, medium, hard). Hard tasks (difficulty >= 0.75) trigger coalition formation: a TaskDecomposer splits the task into subtasks, an AuctionEngine runs sealed-bid auctions where specialists bid cheaply and generalists bid expensively, and a CoalitionFormation module elects a coordinator by reputation. The coordinator has read access to all members private latent states. Over 201 total tasks, 92 were handled by coalitions and 109 by solo agents. Final role distribution — code: agents [0, 3, 6, 9, 12, 15, 18], research: agents [1, 4, 7, 10, 13, 16, 17, 19], visual: agents [2, 5, 8, 11, 14]. Evolution fires every 200 ticks: the bottom 10% of agents by token balance are replaced by mutated offspring of the top 10%. A total of 2 replacement events occurred. Shared memory consists of an EpisodicMemory (FAISS-indexed task-solution pairs) and a KnowledgeDistiller that compresses episodic records into a semantic graph via k-means clustering every 100 ticks.

RESULTS

Over the full run, mean solution quality was 0.795. Coalition formation achieved a 100% win rate on hard tasks across all ticks. The episodic memory accumulated 101 records, which the KnowledgeDistiller compressed into 0 semantic nodes. RAG retrieval provided a quality bonus on coalition tasks from tick 1 onward. Role specialisation emerged without label supervision: by tick 101, distinct code, research, and visual clusters were identifiable from fingerprint dot-product similarity alone. The HistorianAgent issued 2 structured self-reports. Final bottlenecks identified: none detected. Recommendations generated: society healthy — no action needed. The SocietyModel trained for 978 steps online with final loss 1.37, distributing curiosity bonuses to agents whose behaviour surprised its predictions.

DISCUSSION

PANTHEON demonstrates that emergent specialisation is sufficient for AGI-level collective problem solving at laptop scale. The key insight is that intelligence is a property of the collective, not of any individual component. A society of 3M-parameter agents with a well-designed economy outperforms pipeline architectures on tasks requiring coordination, specialisation, and memory. The compute-token economy served as a self-regulating coordination mechanism: specialists naturally bid cheaply for domain tasks, routing emerged without hardcoded rules, and evolutionary pressure eliminated underperforming agents without manual intervention. Limitations include: quality saturation at difficulty tiers prevents measurement of marginal RAG improvement; the visual specialist bottleneck persisted across runs, suggesting the economy needs a diversity incentive; and the SocietyModel requires more event diversity to reduce prediction loss below 1.0. Future work: extend to 50+ agents, introduce inter-coalition competition, and replace the mock coherence scorer with a local Llama-3.2-3B-Instruct model for genuine hypothesis quality measurement.

STRUCTURED DEBATE

Topic: Did economic pressure or task exposure drive specialisation more?

> Agent 1 (score=0.763): **task exposure drove specialisation** — Fingerprint updates fire on every successful task, not just at death. Agents accumulate domain bias through repeated success long before economic pressure becomes critical.
>
> Agent 2 (score=0.761): **both forces are inseparable** — Task exposure shapes the fingerprint; the economy determines survival. Neither alone produces stable specialisation. The interaction between them is the mechanism.
>
> Agent 0 (score=0.724): **economic pressure drove specialisation** — Agents that failed to earn tokens were replaced. Survivors were those whose fingerprints matched high-reward tasks. The economy is the primary selection pressure.

FIGURES

[FIG1] Mean quality across historian report intervals: [0.239, 0.799]

[FIG2] Final agent role distribution by fingerprint cluster: {'code': 7, 'research': 8, 'visual': 5}